DatecsPay

card readers communication protocol

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1. Packet format;

For convenience, we will use “card reader” or “pinpad” for indicating the DatecsPay card reader, and “ext device” for indicating the external device, which is communicating with the card reader (this can be a mobile device, tablet, PC and etc).

1.1. EXT DEVICE TO CARD READER:

Example: `3E 3D 00 00 01 00 02`

Syntax	Description	Example
>	Start packet (ASCII symbol '>')	>
CMD	Command number	3D
00	Fixed value	0
LH LL	Length of data	00 01
<DATA>	Data	0
CSUM	XOR of all bytes in the packet	2

1.2. CARD READER TO EXT DEVICE:

Example: `3E 00 00 00 00 3E`

Syntax	Description	Example
>	Start packet (ASCII symbol '>')	>
00	Fixed value	0
ST	Status (see error codes)	0
LH LL	Length of data	00 00
<DATA>	Data	00 00
CSUM	XOR of all bytes in the packet	3E

Note : The timeout between different bytes is 2 seconds.

Once the card reader receives a valid packet, it will return a response to the external device, where the field ST in the response will indicate if the command is executed successfully or if there is any error, which occurred. The external device can set timeout = 5 seconds for awaiting a response from the card reader after any kind of command was sent to the card reader.

Commands, which can be sent to the card reader:

• 0x3D – Borica command

• 0x40 – External interned command

Events, which can be received unexpectedly from the card reader:

• 0x0E – Borica event

• 0x0F – External internet event

Every command can contain multiple subcommands and every event can contain multiple subevents. All subcommands and subevents are packed in the <DATA> field – usually the first byte of the <DATA> field (DATA[0]).

2. Transactions;

A transaction with a payment terminal involves a set of commands for exchanging data between the payment terminal (referred to as "card reader" or "pinpad") and a banking host. This process encompasses various operations such as payment processing, device configuration, card verification, and other related activities. When the pinpad does not have built-in Internet connectivity (such as GPRS or WiFi), an "ext device" (which can be an Android device or another system with Internet connectivity) is used to facilitate the communication with the banking host.

Borica command (0x3D) contains multiple subcommands, which are associated with almost all activities, which are possible to be executed from the card reader, like for example loading configurations, starting transactions, getting information for already started or already finished transactions and etc.

2.1. Subcommand 0x01: TRANSACTION START

Summary format of the transaction and the response from the pinpad.

		Description
Direction EXT DEVICE >>		
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`LH LL`	Length of data
	`[SUBCMD]`	Subcommand
	`<DATA: VAR>`	Data
[Transaction type]:		
	1	PURCHASE
	2	PURCHASE + CASHBACK
(Bulgaria only)	3	PURCHASE + REFERENCE
	4	CASH ADVANCE
	5	AUTHORIZATION
	6	PURCHASE + CODE
	7	VOID OF PURCHASE
	8	VOID OF CASH ADVANCE
	9	VOID OF AUTHORIZATION
	10	END OF DAY
	11	LOYALTY BALANCE

	12	LOYALTY SPEND
	13	VOID OF LOYALTY SPEND
	14	TEST CONNECTION
	15	TMS UPDATE
	16	REFUND (ROMANIA ONLY)
<Transaction parameters: VAR>:		
	TAG_AMOUNT -0x81	[HEX BIG ENDIAN - 4 bytes]
	TAG_CASHBACK-0x9F04	[HEX BIG ENDIAN - 4 bytes]
	TAG_RRN -0xDF01	[STRING]
	TAG_AUTH_ID -0xDF02	[STRING]
	TAG_REF -0xDF03	~ not used
	TAG_TIP -0xDF63	[HEX BIG ENDIAN - 4 bytes]
Direction CARD READER<<		
	0x00	Fixed value '00'
	ST	Status
	LH LL	Length of data

1. 0x01: START PURCHASE

NEEDED TAGS: 0x81

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x08	The data length is 8 bytes.
	0x01	TRANSACTION START subcommand.
	0x01	HEX format, PURCHASE.
	0x81 04 00 00 00 32	AMOUNT (tag 0x81) = 50 (0.50).
	0xBC	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 08 01 01 81 04 00 00 00 32 BC	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

2. 0x01: START PURCHASE + GRATUITY / TIP

NEEDED TAGS: 0x81, 0xDF63

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x0F	The data length is 15 bytes.
	0x01	TRANSACTION START subcommand.
	0x01	HEX format, PURCHASE.
	0x81 04 00 00 00 1E	AMOUNT (tag 0x81) = 30 (0.30).
	0xDF 63 04 00 00 00 0A	GRATUITY / TIP (tag 0xDF63) = 10 (0.10).
	0x25	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 0F 01 01 81 04 00 00 00 1E DF 63 04 00 00 00 0A 25	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

Note: Tag **0x81** should contain the total <**Amount> + <Gratuity>**. If you need to start a transaction for total amount = 1.10 (1.00 amount + 0.10 Gratuity), then tag 0x81 should have value = '00 00 00 6E' and tag **0xDF63** should have value = ' 00 00 00 0A'.

3. 0x02: START PURCHASE + CASHBACK

When a customer makes a purchase with a card, the "START PURCHASE + CASHBACK" transaction allows them to receive additional cash in hand, which is added to the total purchase amount.

NEEDED TAGS: 0x81, 0x9F04

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	BORICA CMD command.
	0x00	Fixed value '00'.
	0x00 0x0F	The data length is 15 bytes.
	0x01	TRANSACTION START subcommand.
	0x02	HEX format, PURCHASE + CASHBACK.
	0x81 0x04 0x00 0x00 0x00 0x32	AMOUNT (tag 0x81) = 50 (0.50).
	0x9F04 0x04 0x00 0x00 0x00 0x14	CASHBACK (tag 0x9F04) = 20 (0.20).
	0x33	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 0F 01 02 81 04 00 00 00 32 9F 04 04 00 00 00 14 33	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

Note: Tag 0x81 should contain only the <**Amount**>. If you need to start a transaction for total amount = 1.10 (1.00 amount + 0.10 Cashback), then tag 0x81 should have value = '00 00 00 64' and tag **0x9F04** should have value = '00 00 00 0A'.

4. *0x03: START PURCHASE + REFERENCE* – (ROMANIA 0x01)

This transaction enables the external device to initiate a purchase with the card reader, and it allows for the inclusion of a reference as a unique identifier for each transaction. This reference can be used to track and manage transactions more flexibly within a POS system.

NEEDED TAGS: `0x81`, `0xDF03`

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	BORICA CMD command.
	0x00	Fixed value '00'.
	0x00 0x21	The data length is 15 bytes.
	0x01	TRANSACTION START subcommand.
	0x03 (BULGARIA**)**	HEX format, PURCHASE + REFERENCE.
	0x01 (ROMANIA**)**	HEX format, PURCHASE
	0x81 0x04 0x00 0x00 0x00 0x0A	AMOUNT (tag 0x81) = 01 (0.01).
	DF 03 06 31 32 33 34 35 36	REFERENCE (tag 0xDF03) = 123456
	0x49	Checksum (XOR of all bytes from start to end of data).
Example		
	>> 3E 3D 00 00 11 01 03 81 04 00 00 00 01 DF 03 06 31 32 33 34 35 36 49	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example		
	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

Note: Usage of Reference in Transactions

The inclusion of a reference (tag DF03) in the transaction allows each transaction to be uniquely identified. This is particularly useful for transaction tracking, reconciliation, and reporting purposes. The reference acts as a unique identifier that can be used by the POS system to manage transactions more effectively.

5. 0x04: START CASH ADVANCE

The "START CASH ADVANCE" transaction initiates a process where the cardholder obtains cash directly from their card account. This is typically done at an ATM or bank branch and involves withdrawing cash against the available credit limit on the card. It's essentially borrowing cash from the credit card issuer, which usually incurs higher fees and interest rates compared to regular purchases.

NEEDED TAGS:	0x81

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x08	The data length is 8 bytes.
	0x01	TRANSACTION START subcommand.
	0x04	HEX format, CASH ADVANCE.
	0x81 04 00 00 00 64	AMOUNT (tag 0x81) = 100 (1.00).
	0xEF	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 08 01 04 81 04 00 00 00 64 EF	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

6. *0x05: START AUTHORIZATION*

This transaction doesn't withdraw the money from the account. It only blocks them. After that the money can be returned with **VOID OF AUTHORIZATION**, or they can finally be withdrawed with the transaction **PURCHASE + CODE**. These types of transactions are used in Hotels for example.

NEEDED TAGS:	0x81

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x08	The data length is 8 bytes.
	0x01	TRANSACTION START subcommand.
	0x05	HEX format, AUTHORIZATION.
	0x81 04 00 00 00 64	AMOUNT (tag 0x81) = 100 (1.00).
	0xEE	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 08 01 05 81 04 00 00 00 64 EE	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

7. 0x06: START PURCHASE + CODE

When **AUTHORIZATION** has been done successfully, the Amount of the transaction was blocked, but still not withdrawed from the account. In order to withdraw the blocked Amount, **PURCHASE + CODE** needs to be done. That is why this type of transaction can successfully be done only after successful **AUTHORIZATION** before it. When the AUTHORIZATION is being approved, unique **RRN** and Authorization ID are returned from \ the host. They should be printed on the receipt.

Then when starting **PURCHASE + CODE**, these **RRN** and Authorization ID from the first transaction need to be used in the command for starting a transaction. The amount also needs to be the same as in the **AUTHORIZATION**.

NEEDED TAGS:	0x81, 0xDF01, 0xDF02

Direction	Byte (s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x1B	The data length is 27 bytes.
	0x01	TRANSACTION START subcommand.
	0x06	HEX format, PURCHASE + CODE.
	0x81 04 00 00 00 64	AMOUNT (tag 0x81) = 100 (1.00).
	0xDF 01 07 36 36 38 30 36 33 35	RRN (tag 0xDF01) = "6680635".
	0xDF 02 06 39 32 30 32 31 32	Authorization ID (tag 0xDF02) = "920212".
	0xCE	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 1B 01 06 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 CE	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start

		to end).
Example	`<< 3E 00 00 00 00 3E`	Complete response from card reader to external device.

8. *0x07: START VOID OF PURCHASE*

Voiding a Purchase. The Amount, **RRN** and **Authorization ID**, which are used here, need to be taken from the Purchase, which will be voided. After sending the command, if the PINPAD discovers a transaction, saved in the memory, which has the same Amount, **RRN** and **Authorization ID** as those, which are sent in the command, the CARD READER will send the Purchase for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server. This way it can void Purchase from previous days.

NEEDED TAGS: `0x81`, `0xDF01`, `0xDF02`

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet.
	`0x3D`	Borica command.
	`0x00`	Fixed value '00'.
	`0x00 0x1B`	The data length is 27 bytes.
	`0x01`	TRANSACTION START subcommand.
	`0x07`	HEX format, VOID OF PURCHASE.
	`0x81 04 00 00 00 64`	AMOUNT (tag 0x81) = 100 (1.00).
	`0xDF 01 07 36 36 38 30 36 33 35`	RRN (tag 0xDF01) = "6680635".
	`0xDF 02 06 39 32 30 32 31 32`	Authorization ID (tag 0xDF02) = "920212".
	`0xCF`	Checksum (XOR of all bytes from start to end of data).
Example	`>> 3E 3D 00 00 1B 01 07 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 CF`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)	`0x3E`	Start packet.
	`0x00`	Fixed value '00'.
	`0x00`	No error (ERROR CODES)
	`0x00 0x00`	The data length is 0 bytes.
	`0x3E`	Checksum (XOR of all bytes

		from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

9. 0x07: START VOID OF PURCHASE + GRATUITY / TIP

Voiding a **Purchase + Gratuity**. The Amount, Gratuity, **RRN** and **Authorization ID**, which are used here, need to be taken from the Purchase + Gratuity, which will be voided. After sending the command, if the CARD READER discovers a transaction, saved in the memory, which has the same Amount, Gratuity, RRN and Authorization ID as those, which are sent in the command, the CARD READER will send the Purchase for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server. This way it can void Purchase from previous days.

NEEDED TAGS:	0x81, 0xDF63, 0xDF01, 0xDF02

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x22	The data length is 34 bytes.
	0x01	TRANSACTION START subcommand.
	0x07	HEX format, VOID OF PURCHASE.
	0x81 04 00 00 00 64	AMOUNT (tag 0x81) = 100 (1.00).
	0xDF 63 04 00 00 00 0A	Gratuity (tag 0xDF63) = 10 (0.10).
	0xDF 01 07 36 36 38 30 36 33 35	RRN (tag 0xDF01) = "6680635".
	0xDF 02 06 39 32 30 32 31 32	Authorization ID (tag 0xDF02) = "920212".
	0x44	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 22 01 07 81 04 00 00 00 64 DF 63 04 00 00 00 0A DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 44	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.

	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

10. 0x07: START VOID OF PURCHASE + CASHBACK

Voiding a **Purchase + Cashback**. The **Amount**, **Cashback**, **RRN** and **Authorization ID**, which are used here, need to be taken from the Purchase + Cashback, which will be voided. After sending the command, if the PINPAD discovers a transaction, saved in the memory, which has the same Amount, **RRN** and **Authorization ID** as those, which are sent in the command, the CARD READER will send the Purchase for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server. This way it can void Purchase from previous days.

NEEDED TAGS: `0x81`, `0x9F04`, `0xDF01`, `0xDF02`

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)	`0x3E`	Start packet.
	`0x3D`	Borica command.
	`0x00`	Fixed value '00'.
	`0x00 0x22`	The data length is 34 bytes.
	`0x01`	TRANSACTION START subcommand.
	`0x07`	HEX format, VOID OF PURCHASE + CASHBACK.
	`0x81 04 00 00 00 32`	AMOUNT (tag 0x81) = 50 (0.50).
	`0x9F 04 04 00 00 00 14`	CASHBACK (tag 0x9F04) = 20 (0.20).
	`0xDF 01 07 36 36 38 30 36 33 35`	RRN (tag 0xDF01) = "6680635".
	`0xDF 02 06 39 32 30 32 31 32`	Authorization ID (tag 0xDF02) = "920212".
	`0x2B`	Checksum (XOR of all bytes from start to end of data).
Example	`>> 3E 3D 00 00 22 01 07 81 04 00 00 00 32 9F 04 04 00 00 00 14 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 2B`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)	`0x3E`	Start packet.
	`0x00`	Fixed value '00'.
	`0x00`	No error (ERROR CODES)

	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

11. 0x08: START VOID OF CASH ADVANCE

Voiding a **Cash Advance**. The **Amount**, **RRN** and **Authorization ID**, which are used here, need to betaken from the Cash Advance, which will be voided. After sending the command, if the CARD taken from the Cash Advance, which will be voided. After sending the command, if the CARD READER discovers a transaction, saved in the memory, which has the same Amount, RRN and Authorization ID as those, which are sent in the command, the CARD READER will send the **Cash Advance** for voiding without waiting for a card.

NEEDED TAGS: `0x81`, `0xDF01`, `0xDF02`

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet.
	`0x3D`	Borica command.
	`0x00`	Fixed value '00'.
	`0x00 0x1B`	The data length is 27 bytes.
	`0x01`	TRANSACTION START subcommand.
	`0x08`	HEX format, VOID OF CASH ADVANCE.
	`0x81 04 00 00 00 64`	AMOUNT (tag 0x81) = 100 (1.00).
	`0xDF 01 07 36 36 38 30 36 33 35`	RRN (tag 0xDF01) = "6680635".
	`0xDF 02 06 39 32 30 32 31 32`	Authorization ID (tag 0xDF02) = "920212".
	`0xC0`	Checksum (XOR of all bytes from start to end of data).
Example	`>> 3E 3D 00 00 1B 01 08 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 C0`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet.
	`0x00`	Fixed value '00'.
	`0x00`	No error (ERROR CODES)
	`0x00 0x00`	The data length is 0 bytes.
	`0x3E`	Checksum (XOR of all bytes from start to end).
Example	`<< 3E 00 00 00 00 3E`	Complete response from card reader to external device.

12. 0x09: START VOID OF AUTHORIZATION

Voiding an **Authorization**. The **Amount**, **RRN** and **Authorization ID**, which are used here, need to be taken from the Authorization, which will be voided. After sending the command, if the CARD READER discovers a transaction, saved in the memory, which has the same **Amount**, **RRN** and **Authorization ID** as those, which are sent in the command, the CARD READER will send the

Authorization for voiding without waiting for a card. But if the CARD READER doesn't discover such a transaction in the memory, it will prompt for a card and after this it will send the data for voiding to the server.

This way it can void Authorization from previous days.

NEEDED TAGS:	0x81, 0xDF01, 0xDF02

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x1B	The data length is 27 bytes.
	0x01	TRANSACTION START subcommand.
	0x09	HEX format, VOID OF AUTHORIZATION.
	0x81 04 00 00 00 64	AMOUNT (tag 0x81) = 100 (1.00).
	0xDF 01 07 36 36 38 30 36 33 35	RRN (tag 0xDF01) = "6680635".
	0xDF 02 06 39 32 30 32 31 32	Authorization ID (tag 0xDF02) = "920212".
	0xC1	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 1B 01 09 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 C1	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.

	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

13. 0x0A: START END OF DAY

Before performing **End Of Day**, Daily Report receipt needs to be printed. It needs to be printed BEFORE the transaction, because if End Of Day is successful, it will delete all transactions from the log, and it will not be possible information for the transactions to be taken anymore. In order to print the receipt, (**04: GET REPORT TAGS**) command needs to be called multiple times. The first call needs to be with parameter (**01 - FIRST RECORD**) and after that it needs to be called in loop with parameter (**02 - NEXT RECORDS**) until the ST field in the response from the CARD READER becomes (08). If ST field from the response has value (08), this means, that the last transaction in the log was reached.

For each transaction the transaction type and some other information needs to be taken, in order to calculate the transaction sums and the card sums (for transactions, different from TEST CONNECTION and END OF DAY, which are successful **(TAG_DF05_TRANSACTION_RESULTvalue = 0**) and which are not reversed transactions (BOR_TAG_TO_TYPE value != 1)).

After performing **End Of Day**, it is recommended to print on the end of the **Daily Report** if the File was sent successfully or not.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x02	The data length is 2 bytes.
	0x01	TRANSACTION START subcommand.
	0x0A	HEX format, END OF DAY.
	None	No transaction parameters.
	0x0A	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 02 01 0A 0A	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).

Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

Note : Before executing the **END_DAY** transaction, ensure to check for any **HANG** or REVERSAL transactions using the **PINPAD STATUS**. If there is a pending **HANG** transaction, it should be added to the report. For this purpose, a test connection (**TEST CONNECTION**) is performed with the Host Server. Thus, the result of the HANG transaction will be included in the daily report.

14. 0x0B: START LOYALTY BALANCE - The command is not used;

NEEDED TAGS:	0x81	Needs to have value 0.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x08	The data length is 8 bytes.
	0x01	TRANSACTION START subcommand.
	0x0B	HEX format, LOYALTY BALANCE.
	0x81 04 00 00 00 00	AMOUNT (tag 0x81) = 0.
	0x84	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 08 01 0B 81 04 00 00 00 00 84	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

15. 0x0C: START LOYALTY SPEND - The command is not used;

NEEDED TAGS: 0x81

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x08	The data length is 8 bytes.
	0x01	TRANSACTION START subcommand.
	0x0C	HEX format, LOYALTY SPEND.
	0x81 0x04 0x00 0x00 0x00 0x3C	AMOUNT (tag 0x81) = 60 points
	0xBF	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 08 01 0C 81 04 00 00 00 3C BF	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

16. 0x0D: START VOID OF LOYALTY SPEND - The command is not used;

NEEDED TAGS: 0x81

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x08	The data length is 8 bytes.
	0x01	TRANSACTION START subcommand.
	0x0C	HEX format, LOYALTY SPEND.
	0x81 0x04 0x00 0x00 0x00 0x3C	AMOUNT (tag 0x81) = 60 points.
	0xBF	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 08 01 0C 81 04 00 00 00 3C BF	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

17. 0x0E: START TEST CONNECTION

After the test is initiated, the card reader establishes a connection with the banking host and sends a request for a test connection. In case of a successful connection and communication, the banking host responds with a confirmation of the successfully established connection.

NEEDED TAGS:	none	
Direction	**Byte(s)**	**Description**
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x02	The data length is 2 bytes.
	0x01	TRANSACTION START subcommand.
	0x0E	HEX format, TEST CONNECTION.
	0x0E	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 02 01 0E 0E	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

Note : Before executing the **END_DAY** transaction, ensure to check for any **HANG** or **REVERSAL** transactions using the **PINPAD STATUS**. If there is a pending **HANG** transaction, it should be added to the report. For this purpose, a test connection (**TEST CONNECTION**) is performed with the Host Server. Thus, the result of the **HANG** transaction will be included in the daily report.

18. 0x0F: START TMS UPDATE

The command is used to initiate the process of updating the internal information of the card reader. This internal information may include software updates, configuration data, or other critical parameters necessary for the proper functioning of the device. Upon receiving the command, the card reader initiates the update process by loading and applying the new information. Upon successful completion of the update, the card reader sends a response to the external device confirming the successful completion of the operation.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet (ASCII symbol '>')
	0x3D	BORICA CMD command
	0x00	Fixed value '00'
	0x00 0x02	The data length is 2 bytes
	0x01	TRANSACTION START subcommand
	0x0F	HEX format, TMS UPDATE
	0x0F	Checksum (XOR of all bytes from start to end)
Example		
	>> 3E 3D 00 00 02 01 0F 0F	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet (ASCII symbol '>')
	0x00	Fixed value '00'
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes
	0x3E	Checksum (XOR of all bytes from start to end)
Example		
	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

Note: The required IP address or domain name of the TMS server, the port number, and any other necessary network configuration details for connecting are typically preloaded or configured beforehand in the initial setup of the system.

19. 0x10: START REFUND – (ROMANIA)

"START REFUND" transaction initiates the process of reimbursing funds from a previous transaction. When a customer requests a refund, this command is sent to the payment terminal to begin the refund process. The refund amount is specified using the AMOUNT tag (tag 0x81).

NEEDED TAGS:	0x81, 0xDF01, 0xDF02

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x1B	The data length is 27 bytes.
	0x01	TRANSACTION START subcommand.
	0x10	HEX format, REFUND.
	0x81 04 00 00 00 64	AMOUNT (tag 0x81) = 100 (1.00).
	0xDF 01 07 36 36 38 30 36 33 35	RRN (tag 0xDF01) = "6680635".
	0xDF 02 06 39 32 30 32 31 32	Authorization ID (tag 0xDF02) = "920212".
	0xC1	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 1B 01 09 81 04 00 00 00 64 DF 01 07 36 36 38 30 36 33 35 DF 02 06 39 32 30 32 31 32 C1	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

3. Subcommands;

In this context, "subcommands" are instructions sent to the pinpad to retrieve data stored in its memory or to configure its settings. For example, through these commands, information about previous transactions can be requested, display language settings can be changed, specific functions of the pinpad can be activated or deactivated, and so on.

3.1. Subcommand 0x00: PING

Ping the card reader. Ping the card reader to verify its availability and functionality.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x01	The data length is 1 byte.
	0x00	PING subcommand.
	0x02	Checksum (XOR of all bytes from start to end of data).
Example	>> 3E 3D 00 00 01 00 02	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES)
	0x00 0x00	The data length is 0 bytes.
	0x3E	Checksum (XOR of all bytes from start to end).
Example	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

3.2. Subcommand 0x02: GET RECEIPT TAGS

At the end of the transaction, this command can be used to get information for the lastly performed transaction. This way a receipt can be printed.

NEEDED TAGS: Tags to be returned	(For reference see "TLV STRUCTURE" and "TAGS" sections below).

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`0x00 0x04`	The data length is 4 bytes
	`0x02`	GET RECEIPT TAGS subcommand
	`<TAGS: VAR>`	Tags to be returned: `0x81 DF 10`
	`0x4B`	Checksum (XOR of all bytes from start to end)
Example : If Purchase was performed and we want to get the amount and the transaction type, then we need to send the following command:		
	`>> 3E 3D 00 00 04 02 81 DF 10 4B`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x00`	Fixed value '00'
	`0x08`	No error (ERROR CODES)
	`0x00 0x00`	The data length is 0 bytes
	`<DATA: VAR>`	Returned tags data
	`0x36`	Checksum (XOR of all bytes from start to end)
Example		
	`<< 3E 00 08 00 00 36`	Complete response from card reader to external device.

3.3. Subcommand 0x25:SELECT CHIP APPLICATION

Sending selected card Application Identifier:

- **Process**: During the processing of the EVENT_SELECT_CHIP_APPLICATION, the identifier of the selected application is sent to the card.
- **Identifier**: In most cases, this identifier is the sequential number of the application in the list from the event.

Restarting the Procedure:

- After sending the command, the payment procedure restarts.
- The card must be presented again to ensure that the correct application is selected and used for the transaction.

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x0A	Data length is 1 byte.
	0x25	Subcommand: `0x25-SELECT PAY APPLICATION`
	<INDEX>	Selected Application index
	0xXX	Checksum (XOR of all bytes).
Example	3E 3D 00 00 01 25 01 36	Complete command from external device to card reader.
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00	Len.
	0x00	Len.
	0x3E	CRC

3.4. Subcommand 0x03: TRANSACTION END

This command is being used if the EXT DEVICE needs externally to force quit the transaction, after it was started, or in the end of every transaction.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`0x00 0x03`	The data length is 3 bytes
	`0x03`	TRANSACTION END subcommand
	`<CFM: 2>`	Confirmation (optional):
		`0x00 0x00` - END FAIL
		`0x00 0x01` - END OK
	`0x03`	Checksum (XOR of all bytes from start to end)
Example		
	`>> 3E 3D 00 00 03 03 00 00 03`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x00`	Fixed value '00'
	`0x00`	No error (ERROR CODES)
	`0x00 0x00`	The data length is 0 bytes
	`0x3E`	Checksum (XOR of all bytes from start to end)
Example		
	`<< 3E 00 00 00 00 3E`	Complete response from card reader to external device.

Note :<CFM: 2>:

This parameter is optional. If it is not present, then the "END OK" is used by default. If it is present, then it should have any of the following values:
[CFM HIGH][CFM LOW]:
00 00 - END FAIL
00 01 - END OK

3.5. Subcommand 0x04: GET REPORT TAGS

This command is being used when we need to get some information for some specific transaction. We can get information for the last transaction (with parameter (**03 - LAST RECORD**)), then we can get information for all transactions one by one before the last (if we call the same command multiple times,but with parameter (**04 - PREVIOUS RECORD**). We can also get information for the first one (**01 -FIRST RECORD**) and then

call the same command multiple times to get information for all next commands one by one (with parameter (**02 - NEXT RECORDS**)). It is similar to the command (**02: GET RECEIPT TAGS**).

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`0x00 0x04`	The data length is 4 bytes
	`0x04`	GET REPORT TAGS subcommand
	`<TYPE: 2>`	Record type:
		`0x00 0x01` - FIRST RECORD
		`0x00 0x02` - NEXT RECORDS
		`0x00 0x03` - LAST RECORD
		`0x00 0x04` - PREVIOUS RECORD
		`0x00 0x10` - GET ALL RECORDS AS EVENTS
	`<TAGS: VAR>`	Tags to be returned. (For reference see "TLV STRUCTURE" and "TAGS" sections below)
Example		
	`>> 3E 3D 00 00 04 04 00 01 81 83`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x00`	Fixed value '00'
	`0x00`	No error (ERROR CODES)
	`0x00 0x06`	The data length is 6 bytes
	`<DATA: VAR>`	Returned tags data:
		`81 04 00 00 00 64` - The first transaction in this device has amount = 100 (1.00).
	`0xD9`	Checksum (XOR of all bytes from start to end)
Example		
	`<< 3E 00 00 00 06 81 04 00 00 00 64 D9`	Complete response from card reader to external device.

3.6. Subcommand 0x05: GET REPORT INFO

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet (ASCII symbol '>')
	0x3D	BORICA CMD command
	0x00	Fixed value '00'
	0x00 0x01	The data length is 1 byte
	0x05	GET REPORT INFO subcommand
	0x07	Checksum (XOR of all bytes from start to end)
Example		
	>> 3E 3D 00 00 01 05 07	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet (ASCII symbol '>')
	0x00	Fixed value '00'
	0x00	No error (ERROR CODES)
	0x00 0x02	The data length is 2 bytes
	<RECCNT: 2>	The count of records:0x01 0x6B - The count of records is 363
	0x56	Checksum (XOR of all bytes from start to end)
Example		
	<< 3E 00 00 00 02 01 6B 56	Complete response from card reader to external device.

3.7. Subcommand 0x06: GET PINPAD INFO

Subcommand 0x06: GET PINPAD INFO is specified under the ISO 8583 standard, which provides a standardized set of commands for POS device communication.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`0x00 0x01`	The data length is 1 byte
	`0x06`	GET PINPAD INFO subcommand
	`0x04`	Checksum (XOR of all bytes from start to end)
Example		
	`>> 3E 3D 00 00 01 06 04`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x00`	Fixed value '00'
	`0x00`	No error (ERROR CODES)
	`0x00 0x2B`	The data length is 43 bytes
	`<DATA: 43 BYTES>`	Data containing model name, serial number, software version, terminal ID, and menu type.
	<MODEL NAME: 20 BYTES>	Model of the PINPAD
	`<SERIAL NUMBER: 10 BYTES>`	Serial number of the PINPAD
	`<SOFT VER: 4 BYTES>`	Software version of the PINPAD
	`<TERMINAL ID: 8 BYTES>`	Terminal ID of the PINPAD
	`[MENU TYPE]`	Type of the menu of the terminal 1-MENU_TYPE_TRADING 2-MENU_TYPE_TRADING_REF 3-MENU_TYPE_TRADING_ CASHBACK 4-MENU_TYPE_HOTEL_RENT_A_CAR 5-MENU_TYPE_BAR_RESTAURANT 6-ENU_TYPE_BANK_CHANGE

	7-MENU_TYPE_TEST
0x45	Checksum (XOR of all bytes from start to end)
Example	
<< 3E 00 00 00 2B 42 6C 75 65 50 61 64 2D 35 30 00 00 00 00 00 00 00 00 00 30 39 30 31 39 30 30 30 30 31 01 01 17 00 39 33 38 30 30 31 33 33 07 45	Complete response from card reader to external device.

Response Clarification:

- '>': 0x3E - Start of the packet.
- 00: 0x00 - Fixed value '00'.
- ST: 0x00 - No error (ERROR CODES)
- LH LL : 0x00 0x2B - The data length is 43 bytes.
- <MODEL NAME: 20 BYTES>: '42 6C 75 65 50 61 64 2D 35 30 00 00 00 00 00 00 00 00 00 00' - STRING format, "BluePad-50"
- <SERIAL NUMBER: 10 BYTES>: '30 39 30 31 39 30 30 30 30 31' - STRING format, "0901900001"
- <SOFT VER: 4 BYTES>: '01 01 17 00' - HEX format, version 1.1.23.0
- <TERMINAL ID: 8 BYTES>: '39 33 38 30 30 31 33 33' - STRING format, "93800133"
- [MENU TYPE]: 0x07 - HEX format, <MENU_TYPE_TEST>
- CSUM: 0x45 - Xor of all bytes (3E 00 00 ..... 33 07).

3.8. Subcommand 0x07: GET RTC

Subcommand 0x07: GET RTC, specified under the ISO 8583 standard, retrieves the current date and time from the PINpad's Real-Time Clock (RTC).

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`0x00 0x01`	The data length is 1 byte
	`0x07`	GET RTC subcommand
	`0x05`	Checksum (XOR of all bytes from start to end)
Example		
	`>> 3E 3D 00 00 01 07 05`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x00`	Fixed value '00'
	`0x00`	No error (ERROR CODES)
	`0x00 0x06`	The data length is 6 bytes
	`year[1]`	Year in HEX format (1 byte)
	`month[1]`	Month in HEX format (1 byte)
	`date[1]`	Date in HEX format (1 byte)
	`hour[1]`	Hour in HEX format (1 byte)
	`min[1]`	Minute in HEX format (1 byte)
	`sec[1]`	Second in HEX format (1 byte)
	`0x29`	Checksum (XOR of all bytes from start to end)
Example		
	`<< 3E 00 00 00 06 13 03 14 09 30 2C 29`	Complete response from card reader to external device.

3.9. Subcommand 0x08: SET RTC

Subcommand 0x08: SET RTC, specified under the ISO 8583 standard, sets the current date and time on the PINpad's Real-Time Clock (RTC).

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`0x00 0x07`	The data length is 7 bytes
	`0x08`	SET RTC subcommand
	`year[1]`	Year in HEX format (1 byte)
	`month[1]`	Month in HEX format (1 byte)
	`date[1]`	Date in HEX format (1 byte)
	`hour[1]`	Hour in HEX format (1 byte)
	`min[1]`	Minute in HEX format (1 byte)
	`sec[1]`	Second in HEX format (1 byte)
	`0x1D`	Checksum (XOR of all bytes from start to end)
Example		
	`>> 3E 3D 00 00 07 08 13 03 14 09 30 2C 1D`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x00`	Fixed value '00'
	`0x00`	No error (ERROR CODES)
	`0x00 0x00`	The data length is 0 bytes
	`0x3E`	Checksum (XOR of all bytes from start to end)
Example		
	`<< 3E 00 00 00 00 3E`	Complete response from card reader to external device.

3.10. Subcommand 0x1A: GET PINPAD STATUS

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet (ASCII symbol '>')
	0x3D	BORICA CMD command
	0x00	Fixed value '00'
	0x00 0x01	The data length is 1 byte
	0x1A	GET PINPAD STATUS subcommand
	0x18	Checksum (XOR of all bytes)
Example		
	>> 3E 3D 00 00 01 1A 18	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet (ASCII symbol '>')
	0x00	Fixed value '00'
	0x00	No error (60)
	0x00 0x02	The data length is 2 bytes
	<REVERSAL>	Shows if any transaction needs to be reversed
		0x00 No transaction needs to be reversed
		'R' (0x52) Transaction needs to be reversed ('R')
		'C' (0x43)Hang transaction ('C')
	<END DAY>	Shows if end of day needs to be performed
		0x00 No end of day required
		0x01 End of day required
	<TMS UPDATE>	0x02 TMS update required Romania only!
	0x3C	Checksum (XOR of all bytes)
Example		
	<< 3E 00 00 00 02 00 00 3C	Complete response from card reader to external device.

3.11. Subcommand 0x24: GET REPORT TAGS BY STAN

This command is being used in case that it is needed a information for some specific transaction to be taken. To specify the transaction, its STAN (transaction number) needs to be used.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x3D`	BORICA CMD command
	`0x00`	Fixed value '00'
	`LH LL`	Length High and Low bytes indicating the data length
	`0x24`	GET REPORT TAGS BY STAN subcommand
	`<STAN: 4>`	Transaction number in big-endian format
	`<TAGS: VAR>`	Tags to be returned
	`CSUM`	Checksum (XOR of all bytes)
Example		
	`>> 3E 3D 00 00 0C 24 00 00 01 46 DF 05 DF 25 DF 60 81 72`	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	`0x3E`	Start packet (ASCII symbol '>')
	`0x00`	Fixed value '00'
	`ST`	Status byte indicating success or error
	`LH LL`	Length High and Low bytes indicating the data length
	`<DATA: VAR>`	Returned data containing requested tags
	`CSUM`	Checksum (XOR of all bytes)
Example		
	`<< 3E 00 00 00 19 DF 05 04 00 00 00 00 DF 25 02 00 01 DF 60 04 00 00 00 03 81 04 00 00 00 64 59`	Complete response from card reader to external device.

Response Clarification:

- **LH LL**: `0x00 0x19` - The data length is 25 bytes.

- **DATA (VAR)**: `'DF 05 04 00 00 00 00 DF 25 02 00 01 DF 60 04 00 00 00 03 81 04 00 00 00 64'` - HEX format, this is the information returned for the transaction with STAN = 326.

  This data contains multiple elements, composed of `<TAG, LEN, VALUE>`:

  - **Transaction Result**:
    - **TAG**: `DF 05`
    - **LEN**: `04`
    - **VALUE**: `00 00 00 00` - Transaction result tag has value 0, which means the transaction was successful.
  - **Payment Interface**:
    - **TAG**: `DF 25`
    - **LEN**: `02`
    - **VALUE**: `00 01` - Payment interface has value 1. This indicates the transaction was made with the contactless interface.
  - **Contactless Card Scheme**:
    - **TAG**: `DF 60`
    - **LEN**: `04`
    - **VALUE**: `00 00 00 03` - Contactless card scheme has value 3. This indicates the transaction was made with a Visa Contactless card.
  - **Transaction Amount**:
    - **TAG**: `81`
    - **LEN**: `04`
    - **VALUE**: `00 00 00 64` - The amount of this transaction is 100 (1.00).
- **CSUM**: `0x59` - XOR of all bytes (`3E 00 00 ..... 00 64`).

Note:

If some of the tags listed in the command (EXT DEVICE -> CARD READER) are missing for this transaction, they will be missing in the response. If all of the tags are missing for this transaction, the response will contain an ST field with the value `errNoData (8)` error.

3.12. Subcommand 0x40: External internet subcommands

The external internet command is used for managing communication between the card reader and the host, particularly when the card reader itself does not have **WiFi** or **GSM** capabilities. In such cases, the card reader relies on an external device to facilitate the communication. The external device must handle the following activities:

1. **Opening a Socket**: Establishing a connection to the host.
2. **Closing a Socket**: Terminating the connection to the host.
3. **Sending Data to the Host**: Transmitting data from the card reader to the host.
4. **Receiving Response from the Host**: Receiving data from the host and forwarding it to the card reader.

For this functionality to operate correctly, the external device must correctly implement the "**External Internet Command**" and "**External Internet Event**" protocols. This involves properly managing the socket connections and ensuring accurate data transmission between the card reader and the host.

Key Points

- **External Device Role**: Acts as an intermediary for internet communication when the card reader lacks direct connectivity.
- **Core Activities**:
  - **Open Socket**: Create a network connection to the host.
  - **Close Socket**: End the network connection.
  - **Send Data**: Transmit data from the card reader to the host.
  - **Receive Data**: Get the response from the host and relay it back to the card reader.
- **Implementation**: Correct handling of external internet commands and events is crucial for seamless communication.

By ensuring these processes are handled correctly, the external device enables the card reader to effectively communicate with the host, ensuring smooth operation of transaction processing and other related activities.

1. Subcommand 0x01: RECEIVE DATA

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet (ASCII symbol '>')
	0x40	EXTERNAL INTERNET command
	0x00	Fixed value '00'
	LH LL	Length High and Low bytes indicating the data length
	0x01	RECEIVE DATA subcommand
	<DATA: VAR>	Data received from the host
	CSUM	Checksum (XOR of all bytes)
Example		
	>> 3E 40 00 00 3C 01 14 03 01 00 01 01 16 03 01 00 30 17 D3 84 B3 F7 DF 99 20 A6 DF E2 59 6D 99 3B 5C 05 15 D7 D9 51 A6 69 79 D9 37 5C 40 92 86 67 3B 19 70 0D B3 69 24 CF B7 7F E6 E9 35 34 7A A1 E9 A0	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet (ASCII symbol '>')
	0x00	Fixed value '00'
	ST	Status byte indicating success or error
	LH LL	Length High and Low bytes indicating the data length
	<DATA: VAR>	Returned data
	CSUM	Checksum (XOR of all bytes)
Example		
	<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

Additional Notes:

- When sending a response from the host to the **CARD READER**, the **EXT DEVICE** needs to analyze the current **MTU** value. By default, the **MTU** is 0x0400 (1024 bytes). This means that if the length of the response from the host is larger than the **MTU**, the **EXT DEVICE** must "split" the response into smaller packets and send each one separately to the **CARD READER.**

- When sending the "**0x01: RECEIVE DATA**" command, the **EXT DEVICE** needs to analyze the ST field from the response returned by the **CARD READER**. If the CARD READER returns the ST field with value 0x26 (errBusy), the **EXT DEVICE** must wait 100 milliseconds and resend the same data. This loop should continue until the **CARD READER** returns the ST field with value 0x00 (errNoErr).

2. Subcommand 0x02: EVENT CONFIRM

This command confirms to the CARD READER whether the EXT DEVICE successfully performed an action, which before that was issued from the CARD READER with any of the corresponding events (for example open socket).

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet (ASCII symbol '>')
	0x40	EXTERNAL INTERNET command
	0x00	Fixed value '00'
	LH LL	Length High and Low bytes indicating the data length
	0x02	**EVENT CONFIRM** subcommand
	[TYPE]	Type of event: 0x01: SOCKET OPEN EVENT; 0x02: SOCKET CLOSE EVENT; 0x03: SEND DATA EVENT;
	[RESULT]	Result of the event 0x00: CONFIRM OK; 0x01: CONFIRM FAIL;
	<MTU EXT DEVICE: 2>	Maximum transmit unit of the external device (if present)
	CSUM	Checksum (XOR of all bytes)
Example		
	>> 3E 40 00 00 03 02 02 00 7D	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet (ASCII symbol '>')
	0x00	Fixed value '00'
	ST	Status byte indicating success or error
	LH LL	Length High and Low bytes indicating the data length
	<DATA: VAR>	Returned data (if any)

CSUM	Checksum (XOR of all bytes)
Example	
<< 3E 00 00 00 00 3E	Complete response from card reader to external device.

DESCRIPTION:

EXT DEVICE:

- **40 00 LH LL [SUBCMD] [TYPE] [RESULT] <MTU EXT DEVICE: 2>**
  - **LH LL:** Length High and Low bytes indicating the data length.
  - **[SUBCMD]:** Specifies the command subtype.
  - **[TYPE]:** Indicates the type of event being confirmed:
    - **[01]:** SOCKET OPEN EVENT – Sent only if the CARD READER previously sent the "01: SOCKET OPEN" event.
    - **[02]:** SOCKET CLOSE EVENT – Sent only if the CARD READER previously sent the "02: SOCKET CLOSE" event.
    - **[03]:** SEND DATA EVENT – Sent only if the CARD READER previously sent the "03: SEND DATA" event.
  - **[RESULT]:** Indicates the outcome of the action:
    - **[00]:** CONFIRM OK
    - **[01]:** CONFIRM FAIL
  - **<MTU EXT DEVICE>:** Specifies the Maximum Transmit Unit (MTU) of the external device, represented by two bytes (MTU HIGH and MTU LOW).

    By default, it is set to 0x0400 (1024 bytes).

CARD READER:

- **00 ST LH LL**
  - **ST:** Status byte indicating success or error.
  - **LH LL:** Length High and Low bytes indicating the data length.

EXAMPLES:

(EXT DEVICE -> CARD READER)

- **>> 3E 40 00 00 03 02 02 00 7D**
  - This is a complete command sent from the external device to the card reader.
  - It confirms a SOCKET CLOSE EVENT (TYPE: 0x02) with a successful outcome (RESULT: 0x00).

- The data length is 3 bytes, and the checksum is calculated as XOR of all bytes.

(CARD READER -> EXT DEVICE)

- **<< 3E 00 00 00 00 3E**

  This is a complete response from the card reader to the external device. It indicates no error (ST: 0x00) and has a data length of 0 bytes. The checksum is calculated as XOR of all bytes.

3. Subcommand 0x03: GET MAX MTU

Maximum transmit unit of the CARD READER. The card reader only reports its max MTU (how much bytes maximum can be transferred at once between EXT DEVICE and the CARD READER using the subcommand “0x01: RECEIVE DATA”.

Direction	Byte(s)	Description
(EXT DEVICE -> CARD READER)		
	0x3E	Start packet (ASCII symbol '>')
	0x40	EXTERNAL INTERNET command
	0x00	Fixed value '00'
	LH LL	Length High and Low bytes indicating the data length
	0x03	GET MAX MTU subcommand
	CSUM	Checksum (XOR of all bytes)
Example		
	>> 3E 40 00 00 01 03 7C	Complete command from external device to card reader.
(CARD READER -> EXT DEVICE)		
	0x3E	Start packet (ASCII symbol '>')
	0x00	Fixed value '00'
	ST	Status byte indicating success or error
	LH LL	Length High and Low bytes indicating the data length
	<MTU PINPAD: 2>	Maximum transmit unit of the card reader
	CSUM	Checksum (XOR of all bytes)
Example		
	<< 3E 00 00 00 02 04 00 38	Complete response from card reader to external device.

DESCRIPTION:

EXT DEVICE -> CARD READER:

- **0x3E:** Start packet (ASCII symbol '>').

- **0x40:** EXTERNAL INTERNET command.
- **0x00:** Fixed value '00'.
- **LH LL:** Length High and Low bytes indicating the data length (0x00 0x01, meaning 1 byte).
- **0x03:** GET MAX MTU subcommand.
- **CSUM:** Checksum, calculated as XOR of all bytes from start to end (in this example, 0x7C).

CARD READER -> EXT DEVICE:

- **0x3E:** Start packet (ASCII symbol '>').
- **0x00:** Fixed value '00'.
- **ST:** Status byte indicating success or error (0x00 indicating no error).
- **LH LL:** Length High and Low bytes indicating the data length (0x00 0x02, meaning 2 bytes).
- **<MTU PINPAD: 2>:** Maximum transmit unit of the card reader, 2 bytes (in this example, 0x04 0x00, meaning 0x0400 bytes or 1024 bytes).
- **CSUM:** Checksum, calculated as XOR of all bytes from start to end (in this example, 0x38).

4. Borica Events - 0x0E

Borica Events Overview

At certain points during communication between the external device and the card reader, Borica events are received. These events convey important information related to the transaction being performed or historical transaction data. Different Borica subevents are associated with various aspects of transaction information.

These Borica events and subevents play a crucial role in conveying transaction information between the card reader and the external device, facilitating seamless transaction processing and monitoring.At some point different events can be received from the CARD READER. The Borica event has value 0x0E and the different Borica subevents are mostly connected with information for the transaction, which is currently performed, or a transaction, which was already performed in the past.

4.1. Subevnet 0x01: TRANSACTION COMPLETE

(PAYMENT RESULT) - NO CONFIRMATION NEEDED

- This event signifies the completion of a transaction without requiring any confirmation.
- The card reader sends this event to the external device.
- It includes tags encoded in TLV (Tag-Length-Value) format, providing details such as payment result, payment error, host reference number, host authorization ID, authorized amount, and EMV transaction sequence number.
- Example:

```
(CARD READER -> EXT DEVICE) >> 3E 0E 00 00 1C 01 DF 05 04 00 00 00
00 DF 06 04 00 00 00 00 81 04 00 00 00 00 9F 41 04 00 00 03 67 15
```

Clarification:

(CARD READER -> EXT DEVICE):

- **0x3E:** Start of the packet.
- **EVENT:** Borica event (e.g., 0x0E).
- **00:** Fixed value '00'.
- **LH LL:** Data length indicator.
- **SUBEVENT:** Specific Borica subevent (e.g., 0x01 for TRANSACTION COMPLETE).
- **<DATA: VAR>:** Tags encoded in TLV format providing transaction details.
- **CSUM:** Checksum, calculated as XOR of all bytes from start to end.

0x01: TRANSACTION COMPLETE (PAYMENT RESULT) – NO CONFIRMATION NEEDED

Field	Description
Event:	0E 00 LH LL 01 <DATA: VAR>
<DATA: VAR>:	Tags encoded in TLV format
Tags, encoded in TLV format:	
BOR_TAG_PAY_RES 0xDF05	Result (HEX BIG ENDIAN)
BOR_TAG_PAY_ERR 0xDF06	Error (HEX BIG ENDIAN)
BOR_TAG_HOST_RRN 0xDF07	Host RRN (STRING)
BOR_TAG_HOST_AUTH_ID 0xDF08	Host Auth ID (STRING)
TAG_81_AMOUNT_AUTHORISED_BINARY	Amount Authorized (HEX BIG ENDIAN)
BOR_TAG_EMV_STAN 0x9F41	EMV STAN (BCD)
EXAMPLE:	(CARD READER -> EXT DEVICE) >> 3E 0E 00 00 1C 01 DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 81 04 00 00 00 00 9F 41 04 00 00 03 67 15
CLARIFICATION:	(CARD READER -> EXT DEVICE):
	'>' : 0x3E - Start of the packet
EVENT: 0x0E	Borica event
00: 0x00	Fixed value '00'
LH LL: 0x00 0x1C	The data length is 28 bytes
SUBEVENT: 0x01	TRANSACTION COMPLETE subevent
<DATA: VAR>: '...'	Description of data
CSUM: 0x15	Xor of all bytes

4.2. Subevent 0x02: INTERMEDIATE TRANSACTION COMPLETE

(PAYMENT RESULT) - NO CONFIRMATION NEEDED

- This event is sent when a preceding hung transaction passes just before the current transaction being executed.
- Similar to the **TRANSACTION COMPLETE** event, it includes tags in **TLV** format providing payment result, payment error, host reference number, host authorization ID, authorized amount, and EMV transaction sequence number.

0x02: INTERMEDIATE TRANSACTION COMPLETE

Sent if a hung transaction passed just before the transaction, which is currently being executed.

Field	Description
Event:	0E 00 LH LL 02 <DATA: VAR>
<DATA: VAR>:	Tags encoded in TLV format
Tags, encoded in TLV format:	
BOR_TAG_PAY_RES 0xDF05	Result (HEX BIG ENDIAN)
BOR_TAG_PAY_ERR 0xDF06	Error (HEX BIG ENDIAN)
BOR_TAG_HOST_RRN 0xDF07	Host RRN (STRING)
BOR_TAG_HOST_AUTH_ID 0xDF08	Host Auth ID (STRING)
TAG_81_AMOUNT_AUTHORISED_BINARY	Amount Authorized (HEX BIG ENDIAN)
BOR_TAG_EMV_STAN 0x9F41	EMV STAN (BCD)

4.3. Subevent 0x03: HANG TRANSACTION

This event is triggered when a transaction hangs. Its purpose is to return data about the transaction so that it can be printed as a receipt.

Event Structure:

Field	Description
Event:	0E 00 LH LL 03 <DATA: VAR>
<DATA: VAR>:	Receipts Tags encoded in TLV format

4.4. Subevent 0x3F: EVENT_SELECT_CHIP_APPLICATION

This subevent involves selecting a specific application from multiple applications stored on a chip-enabled payment card.

Multiple Applications on a Chip Card:

- Payment Methods: Credit, debit, contactless payments.
- Access to Services: Banking accounts, loyalty programs.
- Identification and Access Control: ID verification, building access.

User Interaction with the Terminal:

Terminal Prompt: Displays a list of available applications with Subevent 0x3F.

User Selection: The user selects the desired application.

Application Access: The terminal accesses the selected application.

After Selecting the Payment Application:

Sending Application Identifier: The identifier of the selected application is sent to the card reader using COMMAND 0x25: SELECT PAY APPLICATION.

Continuing the Payment Procedure: The payment procedure restarts, requiring the same card to be presented again.

Field	Description
Event:	0E 00 LH LL 3F <DATA: VAR>
<DATA: VAR>:	Tags encoded in TLV format
Tags, encoded in TLV format:	
TAG_0xDFC101	TAG_0xDFC101 Card application name 1 (STRING 1251)
TAG_0xDFC101	TAG_0xDFC101 Card application name 2 (STRING 1251)
….	

4.5. Subevent 0x10: SEND LOG DATA

- This event is triggered as a result of the subcommand "0x04: GET REPORT TAGS" with parameter "00 10 - GET ALL RECORDS AS EVENTS".
- Each record is sent as a separate event.
- It includes data related to log records in TLV format.

Sent as a result of subcommand "0x04: GET REPORT TAGS" with parameter "00 10 - GET ALL RECORDS AS EVENTS".
Every record is sent as a single event.

Field	Description
Event:	0E 00 LH LL 10 <DATA: VAR>
<DATA: VAR>:	Tags encoded in TLV format
Tags, encoded in TLV format:	(No specific tags mentioned)

5. EMV2 Events -0x0B – Subevent: 0x82

EMV2 events, which relate to the card reading procedure and the user interface, cover the interaction between the card reader and the user during a transaction. These events include inserting the card, entering a PIN code, confirming the transaction, and more. Here are some major EMV events and their descriptions:

If the subevent is 0x82, it is related to updating the user interface.

Event Value (HEX)	Event ID	EMV Tags (HEX)	User Action
0x89	EMV2_EVENT_PIN_REQUEST	0x9F34	Enter your PIN
0x00	EMV2_MESSAGE_NOT_WORKING	-	Check the device
0x03	EMV2_MESSAGE_APPROVED	0x8A	Transaction approved
0x07	EMV2_MESSAGE_DECLINED	0x8A	Transaction declined
0x09	EMV2_MESSAGE_PLEASE_ENTER_PIN	0x9F34	Please enter your PIN
0x0F	EMV2_MESSAGE_ERROR_PROCESSING	-	Retry or contact support
0x10	EMV2_MESSAGE_REMOVE_CARD	-	Remove the card from the reader
0x15	EMV2_MESSAGE_PRESENT_CARD	-	Insert or tap the card
0x14	EMV2_MESSAGE_IDLE	-	No action required
0x16	EMV2_MESSAGE_PROCESSING	-	Processing...
0x17	EMV2_MESSAGE_CARD_READ_OK_REMOVE	-	Remove the card from the reader
0x18	EMV2_MESSAGE_TRY_OTHER_INTERFACE	-	Try magnetic stripe or contactless
0x19	EMV2_MESSAGE_CARD_COLLISION	-	Use one card
0x1A	EMV2_MESSAGE_SIGN_APPROVED	-	Sign the transaction
0x1B	EMV2_MESSAGE_ONLINE_AUTHORISATI	0x91	Wait for online

	ON		authorization
0x1C	EMV2_MESSAGE_TRY_OTHER_CARD	-	Use another card
0x1D	EMV2_MESSAGE_INSERT_CARD	-	Insert the card into the reader
0x1E	EMV2_MESSAGE_CLEAR_DISPLAY	-	Clear the device display
0x20	EMV2_MESSAGE_SEE_PHONE	-	Check your phone for instructions
0x21	EMV2_MESSAGE_PRESENT_CARD_AGAIN	-	Re-insert the card into the reader
0x89	EMV2_MESSAGE_ENTER_PIN_ON_PHONE	-	Enter your PIN on the phone
0xFF	EMV2_MESSAGE_NA	-	-
0xF0	EMV2_EVENT_USE_CHIP_READER	-	Use chip reader
0xF1	EMV2_EVENT_READ_OTHER	-	-

6. External internet subevents-0x0F

6.1. External Internet Subevents: SOCKET OPEN 0x01

1) Description

When the external device (EXT DEVICE) receives this subevent, it must open a socket to the specified `<ADDRESS>:<PORT>` with the given `<TIMEOUT>`. If using an APN SIM card, it must use the provided APN, Username, and Password for the SIM card. After setting up the socket, the EXT DEVICE must send the subcommand "**0x02: EVENT CONFIRM**".

CARD READER Packet Structure

Byte(s)	Description
0x0F	External Internet event
00	Fixed value '00'
LH LL	Data length in bytes
Subevent	0x01: SOCKET OPEN subevent
ID	Socket ID (0..255)
Type	Socket type (0x01: TCP, 0x02: UDP, 0x03: TCP TLS, 0x04: UDP TLS)
Address	4 bytes for socket address (0..255 each)
Port	2 bytes for socket port (PORT HIGH, PORT LOW)
Timeout	2 bytes for timeout in seconds (TIMEOUT HIGH, TIMEOUT LOW)
Data	Tags encoded in TLV format (see TLV structure below)

Tags in <DATA>

Tag	Identifier	Format
BOR_TAG_COMM_APN	0xDF50	[STRING]
BOR_TAG_COMM_GPRS_UN	0xDF53	[STRING]
BOR_TAG_COMM_GPRS_PWD	0xDF54	[STRING]
BOR_TAG_HOST_LAN_IP	0xDF6C	[STRING]
BOR_TAG_HOST_LAN_PORT	0xDF6D	[HEX]

2) Example Packet

CARD READER -> EXT DEVICE:

```
3E 0F 00 00 26 01 01 01 55 FF 80 C4 4E ED 00 1E DF 50 06 62 6F 72 69 63 61 DF 53
00 DF 54 00 DF 6C 04 58 CB EF 2E DF 6D 02 4E EE 2A
```

Packet Breakdown

Byte(s)	Description
>: 0x3E	Start of the packet
EVENT: 0x0F	External Internet event
00: 0x00	Fixed value '00'
LH LL: 0x00 0x26	Data length is 38 bytes
SUBEVENT: 0x01	SOCKET OPEN subevent
[ID]: 0x01	Socket ID is 1
[TYPE]: 0x01	TCP
<ADDRESS: 4>: 0x55	0xFF 0x80 0xC4 - Address is 85.255.128.196
<PORT: 2>: 0x4E 0xED	Port is 20205
<TIMEOUT: 2>: 0x00 0x1E	Timeout is 30 seconds
<DATA: VAR>:	'DF 50 06 62 6F 72 69 63 61 DF 53 00 DF 54 00 DF 6C 04 58 CB EF 2E DF 6D 02 4E EE'
	APN (Tag 0xDF50) is "borica", Username (Tag 0xDF53) is "" (none), Password (Tag 0xDF54) is "" (none)
	Host LAN IP (Tag 0xDF6C) is 88.203.239.46, Host LAN Port (Tag 0xDF6D) is 20205
CSUM: 0x2A	Checksum is the XOR of all bytes (3E 0F 00 ..... 4E EE)

Note: **Important:** It's recommended to increase the Socket Timeout specified in this event. Some servers may not respond within the initial timeout period, causing communication failures. For better reliability, choose a timeout value that is at least twice the initially requested timeout. For instance, if the initial timeout is 30 seconds, consider increasing it to 60 seconds.

6.2. External Internet Subevents: SOCKET CLOSE 0x02

1. **Description:** When the external device (EXT DEVICE) receives this subevent, it must close the socket with the specified [ID], which was opened before. Then the EXT DEVICE needs to send the subcommand "**0x02: EVENT CONFIRM**".

2. **CARD READER Packet Structure:**

Byte(s)	Description
0x0F	External Internet event
00	Fixed value '00'
LH LL	Data length in bytes
Subevent	0x02: SOCKET CLOSE subevent
ID	Socket ID (0..255)

3. **Example:**

(CARD READER -> EXT DEVICE) >> `3E 0F 00 00 02 02 01 30`

4. **Clarification:**

(CARD READER -> EXT DEVICE):

- >: 0x3E - Start packet.
- EVENT: 0x0F - EXTERNAL INTERNET event.
- 00: 0x00 - Fixed value '00'.
- LH LL: 0x00 0x02 - The data length is 2 bytes.
- SUBEVENT: 0x02 - SOCKET CLOSE subevent.
- [ID]: 0x01 - ID of the socket = 1.
- CSUM: 0x30 - Xor of all bytes (3E 0F 00 ..... 02 01).

6.3. External Internet Subevents: SEND DATA 0x03

1. **Description:** When the EXT DEVICE receives this subevent, it needs to transmit the <DATA> to the host. Then the EXT DEVICE needs to send the subcommand"**0x02: EVENT CONFIRM**".
2. **CARD READER Packet Structure:**

Byte(s)	Description
0x0F	External Internet event
00	Fixed value '00'
LH LL	Data length in bytes
Subevent	0x03: SEND DATA subevent
ID	Socket ID (0..255)
<DATA>	Data to send to the socket

3. **Example:**

(CARD READER -> EXT DEVICE) >> `3E 0F 00 00 37 03 01 16 03 01 00 30 A9 1C C0 83 3E 74 95 D5 84 3B BE 6E D2 06 DE B2 B4 65 1E 75 CE 43 60 DC 6E 26 45 EE C7 B1 7C 2D 1E A1 E3 ED 9B A1 7F 18 BD 30 24 30 9F C1 75 72`

Clarification:

(CARD READER -> EXT DEVICE):

- >: 0x3E - Start packet.
- EVENT: 0x0F - EXTERNAL INTERNET event.
- 00: 0x00 - Fixed value '00'.
- LH LL: 0x00 0x37 - The data length is 55 bytes.
- SUBEVENT: 0x03 - SEND DATA subevent.
- [ID]: 0x01 - ID of the socket is 1.
- <DATA>: '16 03 01 00 ..... C1 75 72' - The EXT DEVICE doesn't need to understand these bytes; it must just send them to the host.
- CSUM: 0x68 - Xor of all bytes (3E 0F 00 ..... 75 72).

7. ERROR CODES

Error Code ID	Code Value	Description
errNoErr	0	No error
errGeneral	1	Some error
errInvCmd	2	No valid command or subcommand code
errInvPar	3	Invalid parameter
errInvAdr	4	Address is outside limits
errInvVal	5	Value is outside limits
errInvLen	6	Length is outside limits
errNoPermit	7	The action is not permitted in current state
errNoData	8	There is no data to be returned
errTimeOut	9	Timeout has occurred
errKeyNum	10	Invalid key number
errKeyAttr	11	Invalid key attributes (usage)
errKeyUsage	11	Invalid key attributes (usage)
errInvDevice	12	Calling of non-existing device
errNoSupport	13	(Not used in this FW version)
errPinLimit	14	Pin entering limit exceeded
errFlash	15	Some error in flash commands
errHard	16	Some hardware error
errInvCRC	17	(Not used in this FW version)
errCancel	18	The button "CANCEL" is pressed
errInvSign	19	Invalid signature
errInvHead	20	Invalid data in header
errInvPass	21	Incorrect password
errKeyFormat	22	Invalid key format
errSCR	23	Some error in smart card reader
errHAL	24	Error code is returned from HAL functions
errInvKey	25	Invalid key (may be not present)
errNoPinData	26	The PIN length is <4 or >12
errInvRemainder	27	Issuer or ICC key invalid remainder length
errNoInit	28	(Not used in this FW version)
errLimit	29	(Not used in this FW version)
errInvSeq	30	(Not used in this FW version)
errNoPerm	31	The action is not permitted

errNoTMK	32	TMK is not loaded. The action cannot be executed
errInvKek	33	Wrong KEK format
errDubKey	34	Duplicated key
errKBD	35	General keyboard error
errKBDNoCal	36	The keyboard is not calibrated
errKBDBug	37	Keyboard bug detected
errBusy	38	The device is busy, try again
errTampered	39	The device is in security tampered
errEmsr	40	Error in encrypted head
errAccept	41	The button "OK" is pressed
errInvPAN	42	Wrong PAN
errOutOfMemory	43	Not enough memory
errEMV	44	Some EMV error
errCrypt	45	Cryptographic error
errComRcv	46	Something is received
errWrongVer	47	Wrong version. Update to old firmware, for example
errNoPaper	48	Printer is out of paper
errTooHot	49	Printer is overheating
errNoConnected	50	The device is not connected
errUseChip	51	Use the chip reader
errEndDay	52	End the day first

8. TAGS

In the TLV (Tag-Length-Value) format, the tag serves as the identifier of the data and helps in their recognition and interpretation. It determines the type of the data that follows in the TLV format.

Example:

- Tag: 0x9F02
- Length: 0x02
- Value: 0x00000010

Here, the tag 0x9F02 identifies the data as "transaction amount", and the following two bytes (0x02) indicate that the length of the value is two bytes. Then, the value (0x00000010) represents the transaction amount, which is 16 in decimal format.

Tag	Description	HEX	REF.
TAG_DF05_TRANSACTION_RESULT	Transaction Result	0xDF05	REF
TAG_DF06_TRANSACTION_ERROR_CODE	Transaction Error Code	0xDF06	REF
TAG_DF31_TERMINAL_MERCHANT_NAME_BG	Merchant Name (in Bulgarian)	0xDF31	
TAG_DF30_TERMINAL_MERCHANT_CITY_BG	Merchant City (in Bulgarian)	0xDF30	
TAG_DF2F_TERMINAL_MERCHANT_ADDRESS_BG	Merchant Address (in Bulgarian)	0xDF2F	
TAG_DF29_TERMINAL_MERCHANT_POST_CODE	Merchant Post Code	0xDF29	
TAG_DF2E_TERMINAL_MERCHANT_TITLE_BG	Merchant Title (in Bulgarian)	0xDF2E	
TAG_DF28_TERMINAL_MERCHANT_PHONE	Merchant Phone	0xDF28	
TAG_9F1C_TERMINAL_ID	Terminal ID	0x9F1C	
TAG_DF00_CARD_SCHEME	Card Scheme	0xDF00	
TAG_DF0A_MASKED_PAN	Masked PAN	0xDF0A	
TAG_5F20_CARDHOLDER_NAME	Cardholder Name	0x5F20	
BOR_TAG_PAY_INTERFACE	Payment Interface	0xDF25	REF
BOR_TAG_CARD_SCHEME_CL	Card Scheme Classification	0xDF60	REF
TAG_DF10_TRANS_TYPE	Transaction Type	0xDF10	REF
TAG_DF0B_LOYALTY_PTS	Loyalty Points	0xDF0B	
TAG_81_AMOUNT_AUTHORISED	Authorised Amount	0x81	

TAG_DF04_AMOUNT_EUR	Amount in EUR	0xDF04	
TAG_DF23_CVM_SIGNATURE	CVM Signature	0xDF23	REF
TAG_9F41_TRANSACTION_SEQUENCE_COUNTER	Transaction Sequence Counter	0x9F41	
TAG_DF07_HOST_RRN	Host RRN	0xDF07	
BOR_TAG_HOST_AUTH_ID	Host Authorisation ID	0xDF08	
BOR_TAG_HOST_CODE	Host Code	0xDF09	
BOR_TAG_TRANS_BATCH_NUM	Transaction Batch Number	0xDF61	
TAG_9F16_TERMINAL_MERCHANT_ID	Terminal Merchant ID	0x9F16	
BOR_TAG_INTERFACE_ID	Interface ID	0xDF62	
TAG_9F06_TERM_AID	Terminal AID	0x9F06	
TAG_9F26_APP_CRYPTOGRAM	Application Cryptogram	0x9F26	
TAG_9A_TRANSACTION_DATE	Transaction Date	0x9A	REF
TAG_9F21_TRANSACTION_TIME	Transaction Time	0x9F21	REF
BOR_TAG_ISSUER_ID	Issuer ID	0xDF79	REF
BOR_TAG_TO_TYPE	TO Type	0xDF12	REF
BOR_TAG_TERM_CURR_NAME	Terminal Currency Name	0xDF27	
BOR_CMD_TAG_CASHBACK	Cashback Command	0x9F04	
BOR_TAG_BCARD_CID_1	BCARD CID 1	0xDF7D	
BOR_TAG_BCARD_CID_2	BCARD CID 2	0xDF7E	
BOR_TAG_IS_PIN_ENTERED	PIN Entered	0xDF7F	

9. REFERENCES

1. Table: TRANSACTION_RESULTS

TRANSACTION_RESULTS ID*	Description	Code
PAY_TRANSACTION_APPROVED	Transaction approved	0
PAY_TRANSACTION_DECLINED	Transaction declined	1
PAY_TRANSACTION_ERROR	Transaction error	2
PAY_TRANSACTION_TRY_OTHER_IF	Try other interface. Contactless only.	3
PAY_TRANSACTION_TRY_AGAIN	Try again. Contactless only.	4

2. Table: INTERFACES

INTERFACES ID*	Description	Code
PAY_INTERFACE_CHIP	Chip interface used	0
PAY_INTERFACE_CONTACTLESS	Contactless interface used	1
PAY_INTERFACE_MAG_STRIPE	Magnetic stripe interface used	2
PAY_INTERFACE_MANUAL_ENTRY	Manual card entry used	3

3. Table: CL CARD SCHEMES

Contactless Card Scheme ID*	Description	Code
PAY_CL_CARD_SCHEME_VISA_AP	Visa Contactless	0x01
PAY_CL_CARD_SCHEME_PAYPASS	PayPass	0x02
PAY_CL_CARD_SCHEME_VISA	Visa	0x03
PAY_CL_CARD_SCHEME_AMEX	American Express	0x04
PAY_CL_CARD_SCHEME_JCB	JCB	0x05
PAY_CL_CARD_SCHEME_DISCOVER	Discover	0x06
PAY_CL_CARD_SCHEME_BCARD	BCARD	0x0A

4. Table: TRANSACTION TYPES

TRANSACTION TYPES ID*	Description	Code
TRANS_TYPE_PURCH	Purchase	1
TRANS_TYPE_PURCH_CASH	Purchase with Cash	2
TRANS_TYPE_PURCH_REF	Purchase Reference	3

TRANS_TYPE_CASH	Cash Withdrawal	4
TRANS_TYPE_AUTH	Authorization	5
TRANS_TYPE_PURCH_CODE	Purchase with Code	6
TRANS_TYPE_VOID_PURCH	Void Purchase	7
TRANS_TYPE_VOID_CASH	Void Cash	8
TRANS_TYPE_VOID_AUTH	Void Authorization	9
TRANS_TYPE_END_OF_DAY	End of Day	10
TRANS_TYPE_LOYALTY_BALANCE	Loyalty Balance Inquiry	11
TRANS_TYPE_LOYALTY_SPEND	Loyalty Spend	12
TRANS_TYPE_VOID_LOYALTY_SPEND	Void Loyalty Spend	13
TRANS_TYPE_TEST_CONN	Test Connection	14
TRANS_TYPE_TMS_UPDATE	TMS Update	15
TRANS_TYPE_REFUND	Refund (Romania)	16

5. Table: SIGNATURE TYPES

SIGNATURE TYPES ID*	Description	Code
BOR_CVM_SIGN_CARDHOLDER	Cardholder Signature	1
BOR_CVM_SIGN_MERCHANT	Merchant Signature	2

6. Table: CARD ISSUERS

CARD ISSUERS ID*	Description	Code
BOR_ISS_ID_BORICA	Borica	1
BOR_ISS_ID_VISA	Visa	2
BOR_ISS_ID_MASTERCARD	Mastercard	3
BOR_ISS_ID_MAESTRO	Maestro	4
BOR_ISS_ID_AMEX	American Express	5
BOR_ISS_ID_DINERS_CLUB	Diners Club	6
BOR_ISS_ID_JCB	JCB	7
BOR_ISS_ID_AMEX_PIN	American Express (PIN)	8
BOR_ISS_ID_BCARD	BCARD	9

7. Table : TO TYPES

TO TYPES ID*	Description	Code
BOR_TO_TYPE_NONE	None	0
BOR_TO_TYPE_REVERSAL	Reversal	1
BOR_TO_TYPE_HANG	Hang	2

* Note: The identifiers for these values are named conditionally for better compatibility with software development serving the protocol.

8. TLV Structure

When a transaction is performed, some information about it is saved in the memory of the card reader. This information is an array composed of multiple elements called tags in TLV format. TLV (Tag-Length-Value) contains the name of the tag, the length of the data, and the data itself.

Example: Let's say a purchase was performed with an amount of 1.00 BGN. The TLV format would look like this: `81 04 00 00 00 64`

- `81` - This is the Tag (TAG_81_AMOUNT_AUTHORISED - 0x81)
- `04` - This is the Length of the data. It means the next 4 bytes will be the amount of this transaction.
- `00 00 00 64` - This is the Data for tag 0x81 in HEX format, representing the amount 100 (1.00 BGN).

Detailed Example: If the purchase transaction includes more information saved in memory, it might look like this:

```
DF 10 02 00 01 81 04 00 00 00 64 DF 04 04 00 00 00 33 DF 23 02 00 01 DF 25 02 00
00 DF 61 02 00 73 DF 00 0B 56 69 73 61 20 43 72 65 64 69 74 5F 20 1A 56 49 53 41
20 41 43 51 55 49 52 45 52 20 54 45 53 54 2F 43 41 52 44 20 31 31 DF 62 02 43 31
9F 41 04 00 00 16 39 9F 21 03 09 34 22 9A 03 18 10 19 DF 09 04 00 00 00 00 DF 07
0C 38 32 39 32 30 36 36 38 33 30 33 33 DF 08 06 30 33 30 31 30 33 DF 05 04 00 00
00 00 DF 06 04 00 00 00 00 9F 06 08 A0 00 00 00 03 10 10 02
```

Parsed Data:

Tag	Len	Value	Description
DF 10	2	00 01	Tag 0xDF10 (TAG_DF10_TRANS_TYPE): Transaction type "Purchase" (Reference 4)
81	4	00 00 00 64	Tag 0x81 (TAG_81_AMOUNT_AUTHORISED): Amount is 100 (1.00 BGN)
DF 04	4	00 00 00 33	Tag 0xDF04 (TAG_DF04_AMOUNT_EUR): Amount in EUR is 51 (0.51 EUR)
DF 23	2	00 01	Tag 0xDF23 (TAG_DF23_CVM_SIGNATURE): Cardholder signature needed (Reference 5)
DF 25	2	00 00	Tag 0xDF25 (BOR_TAG_PAY_INTERFACE): Payment interface "Chip" (Reference 2)

DF 61	2	00 73	Tag 0xDF61 (BOR_TAG_TRANS_BATCH_NUM): Batch number 115
DF 00	0B	56 69 73 61 20 43 72 65...	Tag 0xDF00 (TAG_DF00_CARD_SCHEME): Card scheme "Visa Credit"
5F 20	1A	56 49 53 41 20 41 43 51...	Tag 0x5F20 (TAG_5F20_CARDHOLDER_NAME): Cardholder name "VISA ACQUIRER TEST/CARD 11"
DF 62	2	43 31	Tag 0xDF62 (BOR_TAG_INTERFACE_ID): Interface ID "C1"
9F 41	4	00 00 16 39	Tag 0x9F41 (TAG_9F41_TRANSACTION_SEQUENCE_COUNTER): Transaction number 1639
9F 21	3	09 34 22	Tag 0x9F21 (TAG_9F21_TRANSACTION_TIME): Time 09:34:22
9A	3	18 10 19	Tag 0x9A (TAG_9A_TRANSACTION_DATE): Date 19.10.2018
DF 09	4	00 00 00 00	Tag 0xDF09 (BOR_TAG_HOST_CODE): Host error code 0 (successful transaction)
DF 07	0C	38 32 39 32 30 36 36 38...	Tag 0xDF07 (TAG_DF07_HOST_RRN): Unique RRN number "829206683033"
DF 08	6	30 33 30 31 30 33	Tag 0xDF08 (BOR_TAG_HOST_AUTH_ID): Unique Authorization ID "030103"
DF 05	4	00 00 00 00	Tag 0xDF05 (TAG_DF05_TRANSACTION_RESULT): Transaction result 0 (PAY_TRANSACTION_APPROVED)
DF 06	4	00 00 00 00	Tag 0xDF06 (TAG_DF06_TRANSACTION_ERROR_CODE): Transaction error code 0 (no error)
9F 06	8	A0 00 00 00 03 10 10 02	Tag 0x9F06 (TAG_9F06_TERM_AID): Application ID A000000003101002 (VISA card)

To easily understand the TLV data, you can use an online TLV parser, such as this one.

9.2. Example Transaction

This example illustrates the entire process needed to successfully execute a transaction.

1. PING

The EXT DEVICE sends the "0x00: PING" subcommand.

- **EXT DEVICE:** `3E 3D 00 00 01 00 02`
- **CARD READER:** `3E 00 00 00 00 3E`

2. TRANSACTION START

The EXT DEVICE sends the "0x01: TRANSACTION START" subcommand. Here, it is an example of "Test Connection". For other transactions, refer to the specific subcommands.

- **EXT DEVICE:** `3E 3D 00 00 0D 01 0E 9A 03 19 03 26 9F 21 03 15 11 02 1F`
- **CARD READER:** `3E 00 00 00 00 3E`

3. SOCKET OPEN

The CARD READER will open a socket and send the "0x01: SOCKET OPEN" subevent.

- **CARD READER:** `3E 0F 00 00 26 01 01 01 55 FF 80 C4 4E ED 00 1E DF 50 06 62 6F 72 69 63 61 DF 53 00 DF 54 00 DF 6C 04 58 CB EF 2E DF 6D 02 4E EE 2A`

4. EVENT CONFIRM

The EXT DEVICE must confirm the socket was opened successfully by sending the "0x02: EVENT CONFIRM" subcommand.

- **EXT DEVICE:** `3E 40 00 00 05 02 01 00 04 00 7C`
- **CARD READER:** `3E 00 00 00 00 3E`

5. SEND DATA

The CARD READER needs to send data to the server using the "0x03: SEND DATA" subevent.

- **CARD READER:** `3E 0F 00 00 C0 03 01 00 BC 60 03 33 00 00 08 00 20 38 00 00 00 C0 00 15 91 00 00 00 03 67 15 11 31 03 26 39 33 38 30 30 31 33 30 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 00 79 00 03 38 30 30 00 04 38 31 01 22 00 12 38 32 42 6C 75 65 50 61 64 2D 35 30 00 04 39 31 00 00 00 04 39 32 00 00 00 04 39 33 00 00 00 04 39 34 00 00 00 04 39 35 00 00 00 04 39 36 00 00 00 04 39 37 00 00 00 04 39 38 00 00 00 04 39 39 00 00 00 48 94 2D 02 E3 6F 16 09 0C 6D 9F B3 C7 6C 4C 83 03 61 11 03 1D 2A F5 FF 16 A2 7F BC AB 6B 5C E8 3E 4B 68 51 FD 77 AE A4 EF EF 8A 75 4C B3 A4 E6 DB 58 34 24 1A 00 00 00 00 36`

6. EVENT CONFIRM

The EXT DEVICE must confirm the data was sent to the host successfully by sending the "0x02: EVENT CONFIRM" subcommand.

- **EXT DEVICE:** `3E 40 00 00 03 02 03 00 7C`
- **CARD READER:** `3E 00 00 00 00 3E`

7. RECEIVE DATA

The response from the host is received by the EXT DEVICE, which needs to transfer the data to the CARD READER using the "0x01: RECEIVE DATA" subcommand.

- **EXT DEVICE:** `3E 40 00 00 70 01 00 6D 60 00 00 03 33 08 10 20 38 00 00 02 C0 00 05 91 00 00 00 03 67 15 11 52 03 26 30 30 39 33 38 30 30 31 33 30 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 00 48 94 2D 02 E3 6F 16 09 0C 6D 9F B3 C7 6C 4C 83 03 61 11 03 1D 2A F5 FF 16 A2 7F BC AB 6B 5C E8 3E 4B 68 51 FD 77 AE A4 EF EF 8A 75 4C B3 A4 E6 DB F5 7D E1 C3 00 00 00 00 B9`
- **CARD READER:** `3E 00 00 00 00 3E`

8. SOCKET CLOSE

The CARD READER sends the "0x02: SOCKET CLOSE" subevent.

- **CARD READER:** `3E 0F 00 00 02 02 01 30`

9. EVENT CONFIRM

The EXT DEVICE must confirm the socket was closed successfully by sending the "0x02: EVENT CONFIRM" subcommand.

- **EXT DEVICE:** `3E 40 00 00 03 02 02 00 7D`
- **CARD READER:** `3E 00 00 00 00 3E`

10. TRANSACTION COMPLETE (PAYMENT RESULT)

The CARD READER sends the "0x01: TRANSACTION COMPLETE (PAYMENT RESULT)" subevent.

- **CARD READER:** `3E 0E 00 00 1C 01 DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 81 04 00 00 00 00 9F 41 04 00 00 03 67 15`

-

11. GET RECEIPT TAGS

The EXT DEVICE sends the "0x02: GET RECEIPT TAGS" command to get information needed to print a receipt. For "Test Connection" there is no receipt, but here is an example for a "Purchase + TIP" transaction.

- **EXT DEVICE:** `3E 3D 00 00 5B 02 81 9F 04 9A 9F 21 9F 06 9F 26 9F 1C 9F 16 5F 2A 9F 41 5F 20 DF 00 DF 01 DF 02 DF 03 DF 04 DF 05 DF 06 DF 07 DF 08 DF 09 DF 0A DF 0B DF 10 DF 12 DF 19 DF 23 DF 25 DF 24 DF 26 DF 27 DF 28 DF 29 DF 2A DF 2B DF 2C DF 2D DF 2E DF 2F DF 30 DF 31 DF 60 DF 61 DF 62 DF 63 DF 64 18`
- **CARD READER:** `3E 00 00 01 42 81 04 00 00 00 1E 9A 03 19 03 20 9F 21 03 18 22 22 9F 06 07 A0 00 00 00 03 20 10 9F 26 08 5D 58 40 B4 E2 51 EE 9B 9F 1C 08 39 33 38 30 30 31 33 30 9F 16 0F 39 39 39 33 39 30 30 30 30 31 30 30 30 30 30 5F 2A 02 09 75 9F 41 04 00 00 03 65 5F 20 10 58 58 58 58 58 58 58 20 20 58 58 58 58 58 58 58 DF 00 0D 56 69 73 61 20 45 6C 65 63 74 72 6F 6E DF 04 04 00 00 00 0F DF 05 04 00 00 00 00 DF 06 04 00 00

This is an example of the entire process that is needed to successfully execute a transaction.
1. The EXT DEVICE sends "0x00: PING" subcommand.
>> (EXT DEVICE): 3E 3D 00 00 01 00 02
<< (CARD READER): 3E 00 00 00 00 3E
2. The EXT DEVICE sends "0x01: TRANSACTION START" subcommand. This is an example for "Test Connection", but if you want to start a different transaction, you can see the examples, given in the subcommand.
>> (EXT DEVICE): 3E 3D 00 00 0D 01 0E 9A 03 19 03 26 9F 21 03 15 11 02 1F
<< (CARD READER): 3E 00 00 00 00 3E
3. At some point the CARD READER will need to open a socket. Then it will send the subevent "0x01: SOCKET OPEN".
<< (CARD READER): 3E 0F 00 00 26 01 01 01 55 FF 80 C4 4E ED 00 1E DF 50 06 62 6F 72 69 63 61 DF 53 00 DF 54 00 DF 6C 04 58 CB EF 2E DF 6D 02 4E EE
2A
4. The EXT DEVICE must confirm if it opened the socket successfully. It must send the subcommand "0x02: EVENT CONFIRM".
>> (EXT DEVICE): 3E 40 00 00 05 02 01 00 04 00 7C
<< (CARD READER): 3E 00 00 00 00 3E
5. Then the CARD READER needs to send some data to the server. The CARD READER sends the subevent "0x03: SEND DATA".
<< (CARD READER): 3E 0F 00 00 C0 03 01 00 BC 60 03 33 00 00 08 00 20 38 00 00 00 C0 00
15 91 00 00 00 03 67 15 11 31 03 26 39 33 38 30 30 31 33 30 39 39 39 33
39 30 30 30 30 31 30 30 30 30 30 00 79 00 03 38 30 30 00 04 38 31 01
22 00 12 38 32 42 6C 75 65 50 61 64 2D 35 30 00 04 39 31 00 00 00 04
39 32 00 00 00 04 39 33 00 00 00 04 39 34 00 00 00 04 39 35 00 00 00

04 39 36 00 00 00 04 39 37 00 00 00 04 39 38 00 00 00 04 39 39 00 00
00 48 94 2D 02 E3 6F 16 09 0C 6D 9F B3 C7 6C 4C 83 03 61 11 03 1D
2A F5 FF 16 A2 7F BC AB 6B 5C E8 3E 4B 68 51 FD 77 AE A4 EF EF
8A 75 4C B3 A4 E6 DB 58 34 24 1A 00 00 00 00 36

6. The EXT DEVICE must confirm if it sent the data to the host successfully. It must send the subcommand "0x02: EVENT CONFIRM".

>> (EXT DEVICE): 3E 40 00 00 03 02 03 00 7C

<< (CARD READER): 3E 00 00 00 00 3E

7. Then the response from the host will be received from the EXT DEVICE and it needs to transfer the data to the CARD READER. The EXT DEVICE must call the subcommand "0x01: RECEIVE DATA".

>> (EXT DEVICE): 3E 40 00 00 70 01 00 6D 60 00 00 03 33 08 10 20 38 00 00 02 C0 00 05
91 00 00 00 03 67 15 11 52 03 26 30 30 39 33 38 30 30 31 33 30 39 39 39
33 39 30 30 30 30 31 30 30 30 30 30 00 48 94 2D 02 E3 6F 16 09 0C 6D
9F B3 C7 6C 4C 83 03 61 11 03 1D 2A F5 FF 16 A2 7F BC AB 6B 5C
E8 3E 4B 68 51 FD 77 AE A4 EF EF 8A 75 4C B3 A4 E6 DB F5 7D E1
C3 00 00 00 00 B9

<< (CARD READER): 3E 00 00 00 00 3E

8. The CARD READER will send the subevent "0x02: SOCKET CLOSE".

<< (CARD READER): 3E 0F 00 00 02 02 01 30

9. The EXT DEVICE must confirm if it closed the socket successfully. It must send the subcommand "0x02: EVENT CONFIRM".

>> (EXT DEVICE): 3E 40 00 00 03 02 02 00 7D

<< (CARD READER): 3E 00 00 00 00 3E

10.The CARD READER sends the subevent "0x01: TRANSACTION COMPLETE (PAYMENT RESULT)".

<< (CARD READER): 3E 0E 00 00 1C 01 DF 05 04 00 00 00 00 DF 06 04 00 00 00 00 81 04 00
00 00 00 9F 41 04 00 00 03 67 15

11.The EXT DEVICE sends "0x02: GET RECEIPT TAGS" command to get information, which is needed to print a receipt for this transaction. In case of "Test Connection" there is no receipt, but the given example is from transaction "Purchase + TIP".

>> (EXT DEVICE): 3E 3D 00 00 5B 02 81 9F 04 9A 9F 21 9F 06 9F 26 9F 1C 9F 16 5F 2A
9F 41 5F 20 DF 00 DF 01 DF 02 DF 03 DF 04 DF 05 DF 06 DF 07 DF
08 DF 09 DF 0A DF 0B DF 10 DF 12 DF 19 DF 23 DF 25 DF 24 DF 26
DF 27 DF 28 DF 29 DF 2A DF 2B DF 2C DF 2D DF 2E DF 2F DF 30
DF 31 DF 60 DF 61 DF 62 DF 63 DF 64 18

<< (CARD READER): 3E 00 00 01 42 81 04 00 00 00 1E 9A 03 19 03 20 9F 21 03 18 22 22 9F
06 07 A0 00 00 00 03 20 10 9F 26 08 5D 58 40 B4 E2 51 EE 9B 9F 1C
08 39 33 38 30 30 31 33 30 9F 16 0F 39 39 39 33 39 30 30 30 30 31 30
30 30 30 30 5F 2A 02 09 75 9F 41 04 00 00 03 65 5F 20 10 58 58 58 58
58 58 58 20 20 58 58 58 58 58 58 58 DF 00 0D 56 69 73 61 20 45 6C 65
63 74 72 6F 6E DF 04 04 00 00 00 0F DF 05 04 00 00 00 00 DF 06 04 00
00 00 00 DF 07 0C 30 30 30 30 30 36 36 39 31 31 33 35 DF 08 06 39 36
39 33 36 34 DF 09 04 00 00 00 00 DF 0A 10 2A 2A 2A 2A 2A 2A 2A 2A
2A 2A 2A 2A 37 38 39 38 DF 10 01 01 DF 12 02 00 00 DF 23 02 00 00
DF 25 02 00 01 DF 26 02 0A 02 DF 27 03 42 47 4E DF 28 07 39 37 30
32 36 31 30 DF 29 04 31 30 30 30 DF 2A 04 54 45 53 54 DF 2E 06 44 41
54 45 43 53 DF 2F 19 42 55 4C 2E 54 5A 41 52 49 47 52 41 44 53 4B 4F
20 53 48 41 4F 53 53 45 45 DF 30 05 53 4F 46 49 41 DF 31 04 54 45 53
54 DF 60 04 00 00 00 03 DF 61 02 00 27 DF 62 02 43 31 DF 63 04 00 00
00 0A DF 64 04 00 00 00 05 CC

12.The EXT DEVICE sends "0x03: TRANSACTION END" command.

>> (EXT DEVICE): 3E 3D 00 00 01 03 01
<< (CARD READER): 3E 00 00 00 00 3E

10. SERVICE COMMAND (Internal use only)

10.1. COMMAND 0x23: CHECK_PASSWORDS;

The coomand used to verify the validity of passwords stored on a device. Upon receiving this command, the device checks the provided passwords against its stored records and returns a response indicating whether the passwords are valid or not.

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0xXX 0xXX	Data length is 0xXX bytes.
	0x23	CHECK_PASSWORDS command.
	0xXX	Password type (1 to 6). 1. MerchantPassword 2. BankPasswords 3. CustomPassword1 4. CustomPassword2 5. CustomPassword3 6. InternalPassword
	<DATA: VAR>	Password to be verified.
	0xXX	Checksum (XOR of all bytes).
Example	3E 3D 00 00 06 23 01 30 30 30 30 27	Верификация на MerchantPassword;
	3E 3D 00 00 06 23 01 0x30 30 30 30 27	Complete command from external device to card reader.
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00 0x00	Data length is 2 byte.
	0x3E	Checksum (XOR of all bytes).

10.2. COMMAND 0x20: EDIT COMMUNICATIONS SETTING

The command for the initial setup and editing of communication parameters on a payment terminal, which are essential for its proper functioning. These parameters are automatically updated during TMS updates. Reading these parameters is done with the GET_MENU_INFO command.

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0xXX 0xXX	Data length is 0xXX bytes.
	0x20	EDIT COMMUNICATIONS SETTING subcommand.
	<PARAMS>	Communication parameters.
	0xXX	Checksum (XOR of all bytes).
	3E 3D 00 00 50 20 DF 80 03 09 30 31 32 33 34 33 32 31 30 DF 51 0E 38 35 2E 32 35 35 2E 31 32 38 2E 31 39 36 DF 52 02 4E ED DF 7B 0E 31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 DF 7C 02 4E EE DF 65 0E 31 39 33 2E 34 31 2E 31 39 30 2E 31 36 39 DF 66 02 1F 40 59	
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00 0x00	Data length is 1 byte.
	0xXX	Checksum (XOR of all bytes).

Communication parameters.

Tag ID	Description	Value
TAG_DF8003_PASSWORD	Internal Password	0xDF8003
TAG_DF51_HOST_IP	Authentication Host IP	0xDF51
TAG_DF52_HOST_PORT	Authentication Host Port	0xDF52
TAG_DF7B_KEYS_IP	Keys Host IP (Key Generation Server)	0xDF7B
TAG_DF7C_KEYS_PORT	Keys Host Port (Key Generation Server)	0xDF7C

TAG_DF65_TMS_IP	TMS Host IP	0xDF65
TAG_DF66_TMS_PORT	TMS Host Port	0xDF66

10.3. COMMAND 0x15: SET CUSTOM TAG (SET MODE TLS)

The command configures custom data tags on a chip-enabled device, specifically establishing Transport Layer Security (**TLS**) mode for secure communication between the device and external systems. This command is used to change the encryption settings of the connection with the host server, whether it is in the **Card Reader**, in the external application software, or via **GPRS**;

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0xXX 0xXX	Data length is 0xXX bytes.
	0x15	SET TAG subcommand.
	0x27,0xA6,0xA6, 0x00	Sequence is part of a communication protocol;
	0xDF 3C 04 00 00 00 00 0D	(Tag 0xDF3C MODE TLS) 0x0B – GPRS MODE 0x0F - internal TLS; 0x0D- external TLS;
Example	3E 3D 00 00 09 15 27 A6 A6 00 DF 3C 01 0D D7	Complete command from external device to card reader.
	0xD7	Checksum (XOR of all bytes).
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00 0x00	Data length is 2 bytes.
	0x3E	Checksum (XOR of all bytes).

10.4. COMMAND 0x27: GET CUSTOM TAG (GET MODE TLS)

Retrieves custom data tags from a Card Reader, enabling access to application-specific information such as communication details or data. This command is used to read the encryption settings of the connection with the host server, whether it is in the Card Reader, in the external application software, or via GPRS;

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0xXX 0xXX	Data length is 0xXX bytes.
	0x27	GET CUSTOM TAG subcommand.
Example	>>3E 3D 00 00 04 27 DF 3C 00 C3	Complete command from external device to card reader.
	0xC3	Checksum (XOR of all bytes).
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00 0x00	Data length
	<DATA>	
	0x3E	Checksum (XOR of all bytes).
Example	<<3E 00 00 00 04 DF 3C 01 0F D7	(Tag 0xDF3C MODE TLS) 0x0B – GPRS MODE 0x0F - internal TLS; 0x0D- external TLS;

10.5. COMMAND 0x21: LOAD KEYS (Initialize keys)

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x02	Data length is 2 bytes.
	0x21	Subcommand: 0x21 -LOAD KEYS
	<TYPE: 1 byte>	1. INIT_KEYS 2. INIT_KEYS_REMOTE 3. UPDATE_KEYS 4. UPDATE_KEYS_REMOTE 5. DECOMMISSION_KEYS 6. DECOMMISSION_KEYS_REMOTE
Example:	3E 3D 00 00 02 21 01 21	
	0x21	Checksum (XOR of all bytes).
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00	Len.
	0x00	Len.
	0x39	Checksum (XOR of all bytes).

10.6. COMMAND 0x1E,0x1F: READ KEYS LIST

Retrieves the list of cryptographic systems keys stored on a device.

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x01	Data length is 1 byte.
	0x1E or 0x1F	Subcommand: 0x1E-READ PUBLIC KEYS LIST; 0x1F -READ SYMMETRIC KEYS;
	0x1C	Checksum (XOR of all bytes).
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00 <Length>	Data length (variable, depending on the number and size of public keys).
	<PublicKeys>	List of public keys. Each key can have a fixed length or be separated by a specific delimiter. The received bytes, indicative of response data, need to be transformed into a string using the 1251 encoding.
	0xXX	Checksum (XOR of all bytes).

10.7. COMMAND 0x0C: CLEAR REVERSAL

Clears the flag in the terminal indicating the presence of a transaction for which no confirmation of receiving the result has been obtained.

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x0A	Data length is 10 byte.
	0x0C	Subcommand: `0x0C-CLEAR_REVERSAL`;
	0x30,0x31,0x32,0x33,0x34, 0x33,0x32,0x31,0x30,0x31	Internal Password: 012343210
	0x31	Checksum (XOR of all bytes).
Example	3E 3D 00 00 0A 0C 30 31 32 33 34 33 32 31 30 31	Complete command from external device to card reader.
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00	Len.
	0x00	Len.
	0x3E	CRC

10.8. COMMAND 0x0B : CLEAR BATCH

Clears the terminal buffers accumulated for the day.

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x0A	Data length is 10 byte.
	0x0B	Subcommand: `0x0B-CLEAR BATCH`;
	0x30,0x31,0x32,0x33,0x34, 0x33,0x32,0x31,0x30,0x31	Internal Password: 012343210
	0x31	Checksum (XOR of all bytes).
Example	3E 3D 00 00 0A 0B 30 31 32 33 34 33 32 31 30 36	Complete command from external device to card reader.
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00	Len.
	0x00	Len.
	0x3E	CRC

10.9. COMMAND 0x1D : READ MENU INFO

Retrieves information about the capabilities and data stored on a device, often utilized in organizing menu structures and options. .

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x3D	Borica command.
	0x00	Fixed value '00'.
	0x00 0x01	Data length is 1 byte.
	0x1D	Subcommand: `0x1D-READ MENU INFO`
	0x1F	Checksum (XOR of all bytes).
Example	3E 3D 00 00 01 1D 1F	Complete command from external device to card reader.
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00	Len.
	0x7B	Len.
	<DATA>	Returned data of menu information
	0x3E	CRC
Example:	3E 00 00 00 7B 31 36 32 33 39 30 30 31 30 39 50 38 30 30 30 30 30 33 39 38 38 30 30 30 30 30 30 36 00 00 00 00 00 00 01 01 22 03 4A 75 6C 20 31 39 20 32 30 32 33 00 00 00 00 00 31 35 3A 32 32 3A 31 38 00 00 38 35 2E 32 35 35 2E 31 32 38 2E 31 39 36 00 00 4E ED 31 39 33 2E 34 31 2E 31 39 30 2E 31 37 30 00 00 4E EE 31 39 33 2E 34 31 2E 31 39 30 2E 31 36 39 00 00 1F 40 03 33 06 21 02 07	Complete response from card reader to external device.

Data returned of COMMAND 0x1D : READ MENU INFO

Field	Size (bytes)	Description	ASCII Representation
PINPAD_SERIAL_NUMBER	10	Serial number of the Pinpad	1623900109
TERMINAL_ID	8	Terminal identifier	P8000003
MERCHANT_ID	16	Merchant identifier	9880000006
SOFTWARE_VERSION	4	Software version of the Pinpad	1.1.34.3
BUILD_DATE	16	Build date of the software	Jul 19 2023 15:22:18
AUTHORIZATION_IP	16	IP address for authorization	85.255.128.196:20205
KEY LOAD IP	16	IP address for keys	193.41.190.170:20206
TMS_IP	16	IP address for Terminal Management System	193.41.190.169:8000
AUTH_NII_HOST	2	Host NII for authorization	333
KMC_NII_HOST	2	Host NII for KMC	621
SYSTEM_KEYS_STATE	1	System key state	KEK_AND_MAC_OK
CURRENCY_NAME	3	System currency name	BGN

10.10. COMMAND 0x0C:RESTART PINPAD

Perferom soft-restart.

Direction	Byte(s)	Description
External Device to Card Reader		
	0x3E	Start packet.
	0x36	Service Internal command.
	0x00	Fixed value '00'.
	0x00 0x01	Data length is 1 byte.
	0x0C	Subcommand: `0x0C:RESTART PINPAD`;
	0x05	Checksum (XOR of all bytes).
Example	3E 36 00 00 01 0C 05	Complete command from external device to card reader.
Card Reader to External Device		
	0x3E	Start packet.
	0x00	Fixed value '00'.
	0x00	No error (ERROR CODES).
	0x00	Len.
	0x00	Len.
	0x3E	CRC

10.11. COMMAND 0x27: GET TERMINAL ID LIST

Retrieve a list of Terminal IDs configured in the terminal.

Direction: **External Device to Card Reader**

Byte(s)	Description
0x3E	Start packet.
0x36	Service internal command.
0x00	Fixed value '00'.
0x00 0x01	Data length is 1 byte.
0x27	Subcommand: 0x27: GET TERMINAL ID LIST.
0x1A	Checksum (XOR of all bytes).

Example Command:

```
3E 36 00 00 01 27 1A
```

Complete command from external device to card reader.

Direction: **Card Reader to External Device**

Byte(s)	Description
0x3E	Start packet.
0x00	Fixed value '00'.
0x00	No error (ERROR CODES).
0xXX	Length of the data (high byte).
0xYY	Length of the data (low byte).
Data	TLV encoded Terminal IDs (9F1C).
0xZZ	**CRC (checksum of all bytes).**

Response Example:

```
3E 00 00 00 12 9F1C08 12345678 9F1C08 87654321 ZZ
```

Where ZZ is the CRC.

10.12. COMMAND 0xXX: SET CURRENT CONFIGURATION

Set the active configuration of the terminal using a specified Terminal ID (TID).

Direction: External Device to Card Reader

Byte(s)	Description
`0x3E`	Start packet.
`0x36`	Service internal command.
`0x00`	Fixed value '00'.
`0x00 0xYY`	Data length in bytes (YY is the TLV length).
`0xXX`	Subcommand: `COMMAND_SET_CURRENT_CONFIG`.
`TLV Data`	Encoded Terminal ID using `TAG 9F1C`
`0xZZ`	**Checksum (XOR of all bytes).**

Direction: Card Reader to External Device

Byte(s)	Description
`0x3E`	Start packet.
`0x00`	Fixed value '00'.
`0x00`	No error (ERROR CODES).
`0x00`	Length of the data (high byte).
`0x00`	Length of the data (low byte).
`0x3E`	Checksum (CRC).
